Brandon Howe

CS + Math Student | AI Safety Research

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Independent AI Safety Experience

Stated vs. Revealed Value Preferences in LLMs

April 2026

LLM Evals | Value Learning

- Replicated and extended the LitmusValues evaluation across four frontier models, comparing stated rankings of 16 values with preferences revealed through 3,000 AIRiskDilemmas.
- Found that GPT-5.4 Nano's stated and revealed preferences correlated at $\rho = 0.656$, while revealed-preference rankings remained nearly identical across GPT and Claude models.
- Wrote up the replication, steering experiments, and results in a post on my blog.

Mechanistic Interpretability Replications

May - August 2026

Mech. Interp. | Linear Probes

- Reproduced Anthropic's Emotion Concepts results on Gemma-2-2B-IT using difference-in-means probes, finding similar geometric structure on a smaller open-weight model.
- Tested the single-direction refusal result on Gemma-2B through directional ablation, activation addition, and weight orthogonalization.

Persona and Model-Utility Experiments

June - July 2026

Personas | PCA / UMAP

- Investigated whether models encode persona-dependent task utility by analyzing hidden-state representations across roughly 90 personas with PCA and UMAP.
- Found a weak assistant axis across personas, with the first principal component explaining 9% of the variance.

ARENA

March - May 2026

PyTorch | Python | RL / Evals

- Implemented transformers from scratch and completed practical work in mechanistic interpretability, reinforcement learning, and model evaluations.

Work History

Software Intern - Mechanical Solutions, Inc.

May 2022 - Present

C# | WPF | C

- Developed and shipped C#/WPF features for user-facing products, including live analysis and shaft inspection workflows.
- Optimized analysis code paths that improved performance of key functions nearly 10×.

Lead Developer - Touch by Touch

March 2021 - Present

React | Flutter | TypeScript

- Adopted as the official scorekeeping platform of New Jersey's high school fencing league, serving 50+ schools, several NCAA university programs, and thousands of fencers.
- Built and maintained real-time scoring, roster management, and rankings systems across React, Flutter, and TypeScript clients.

Software Intern - Agatha

May 2021 - September 2021

Flutter

- Independently built Agatha Health's Flutter MVP, enabling patients to find doctors and pay for video consultations.

Other Projects

CUDA/MPI-based CNN	2025	*Wrote and trained a CNN from scratch as a course project*	CUDA, MPI
COOL Compiler	2025	*Type checker and x86 code generator for COOL written from scratch in C as a course project*	C, Optimization
FIE Stream Clipper	2024	*Uses computer vision to automatically cut FIE livestreams into individual bouts*	OpenCV, C++

Education

B.S. in CS & Applied Mathematics, New Jersey Institute of Technology

May 2028

- 4.0 GPA, with a full-ride scholarship for all four years as part of the Albert Dorman Honors College.
- Division I athlete on the men's fencing team.